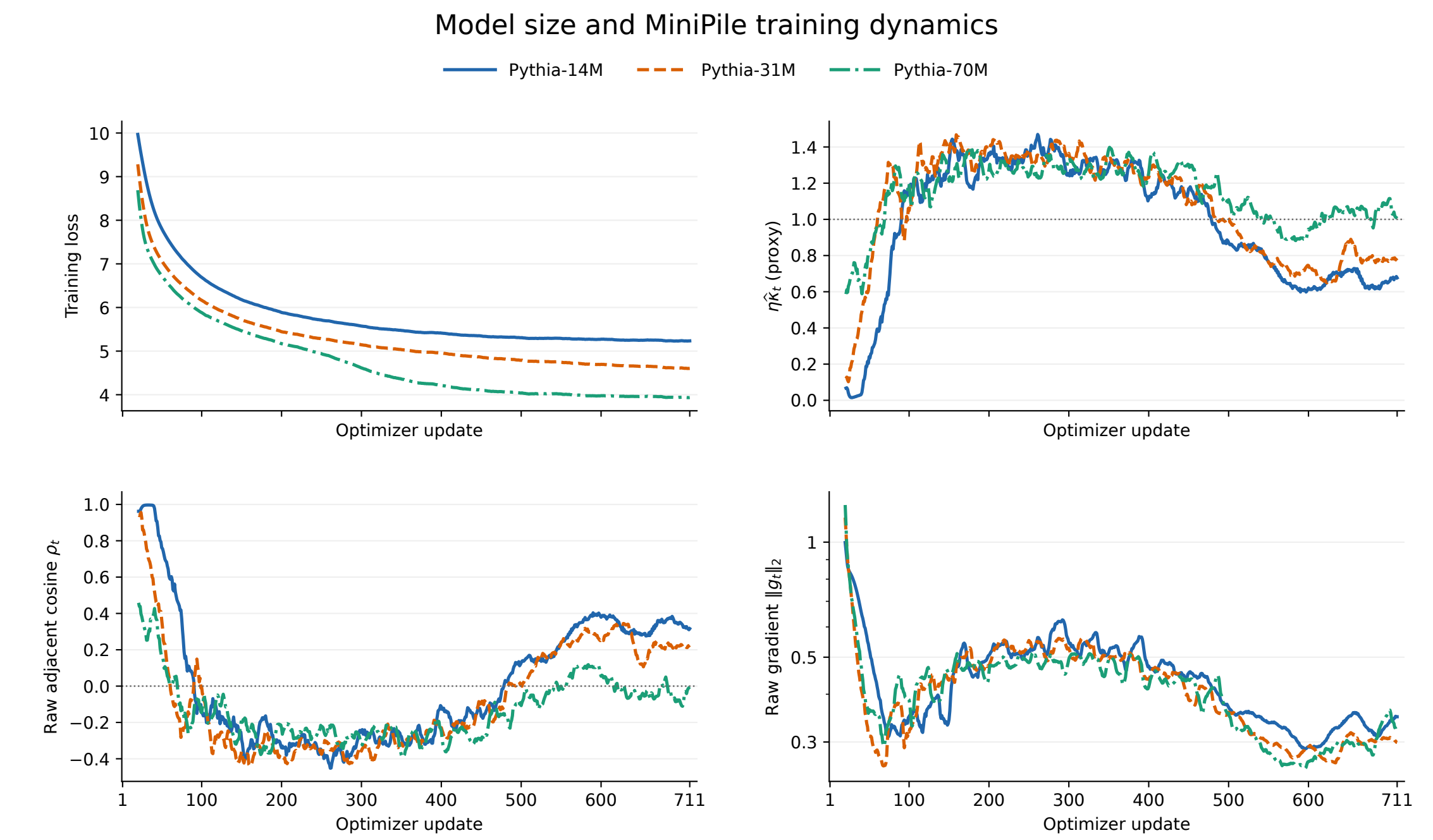

Model size and MiniPile training dynamics
Pythia-14M
Pythia-31M
Pythia-70M
Training loss
Optimizer update
$\hat{\eta\kappa}_t$ (proxy)
Optimizer update
Raw adjacent cosine ρ_t
Optimizer update
Raw gradient $\|g_t\|_2$
Optimizer update
Complete 20-update trailing means; seed 1235. Update 712 excluded before smoothing.
First mean: update 20 (loss/norm), 21 (pairs). Norm axis: log Y. AdamW q is a gradient-change proxy.